1.4 Computer storage

In this lesson

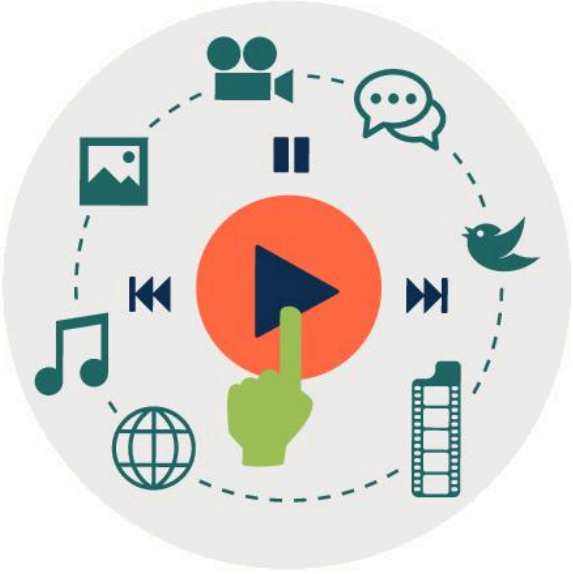
You will learn:

- about computer storage
- that computer storage is an important component of a computer system
- why computer storage is important.

Data files

Different types of media can be created and used on a computer. The media types used on computers include:

- text documents (for example, homework or assignments)
- pictures and photographs
- videos
- music



All types of media used on a computer are stored in files. A file stored on a computer is called a **data file**.

Computer storage

When you create work on a computer, it is important to save it correctly. Work saved in a data file can be used over and over again. If your work is not saved correctly it can be lost.

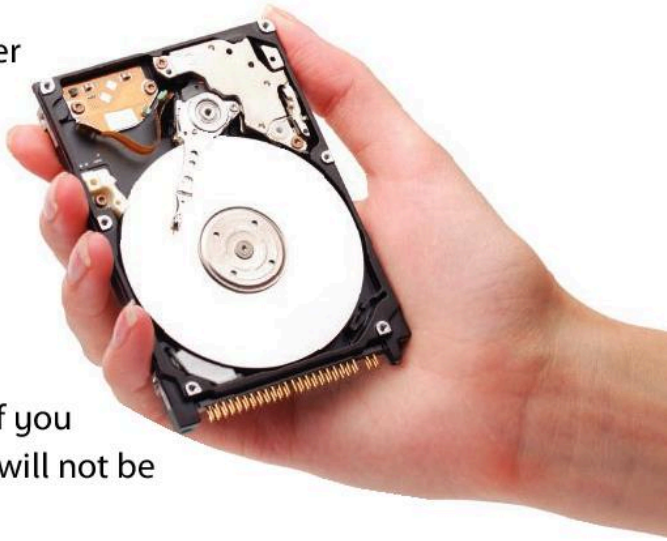
All computers have a place to store data files. It is called computer storage.

Where data files are stored

A computer uses a device called a **storage drive** to save files. Every computer has a storage drive built into the case. A storage drive can save a large amount of information.

For example, a storage drive used in a desktop computer can store:

- 150,000 photographs or
- 8500 hours of music or
- 300,000 books.



Network storage

You will probably not always use the same computer. If you save work onto the storage drive of one computer, you will not be able to use that file if you move to another computer.

Computers in schools and offices are usually connected. We call a group of connected computers a **network**. A network has its own storage drive. Everyone using a network has a special area on the **network storage** drive where they can save their work. If you save your work on your school network drive, you can use it on any computer that is connected to your school network.

Flash drives

A **flash drive** is a small portable storage device. It is small enough to fit in your pocket and can be used to move files from one computer to another. A flash drive is plugged into a computer so that a data file can be saved to it.





Activity

How could you use a flash drive to move homework from a computer at home to a computer at school? Write your answer as a list of steps.



Extra challenge

You have learned that a typical storage drive can hold 8500 hours of music. How many days would it take to play that music? (Hint: there are 24 hours in a day.)

Think again

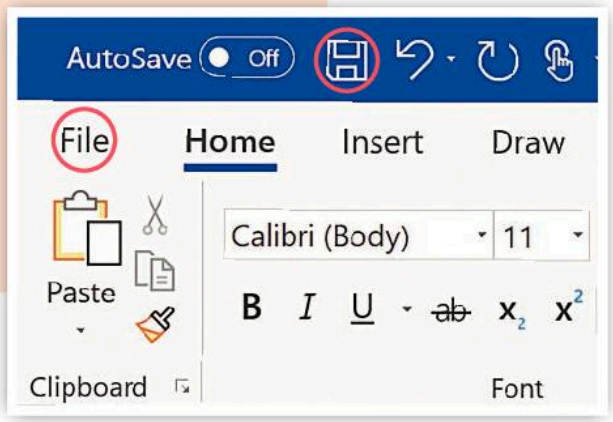
What problems will you have if you don't save your work properly?

1.5 How to use and store your files

In this lesson

You will learn:

- how to use files and folders
- how to store files.



Using and storing files

How to save a file

If you are working in an application such as a word processor, there are two ways to save a file. You can use the File menu or click the Save icon. If your file already has a name, it will be saved with the same name. If it is a new file, you need to enter a file name before you can save the file. You will learn more about choosing a good name for your file in Lesson 1.6.

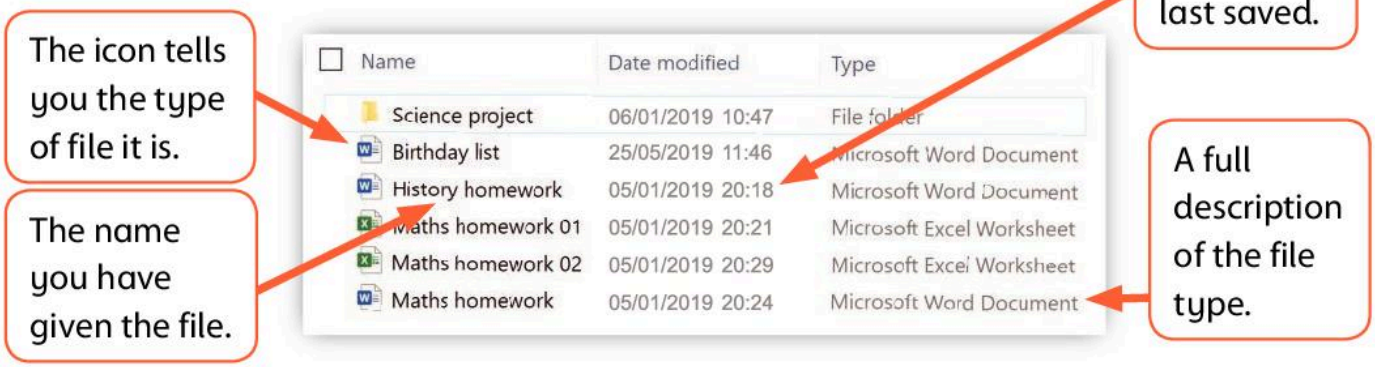
How to copy a file

You can reduce the risk of losing important work by making a copy of the file. The copy is called a **back-up file**.

- Find the file in File Explorer.
- Right click your mouse over the file, then select Copy from the menu that appears.
- Move to the folder where you want to save your back-up. Right click your mouse and select Paste from the menu.

How to find a file

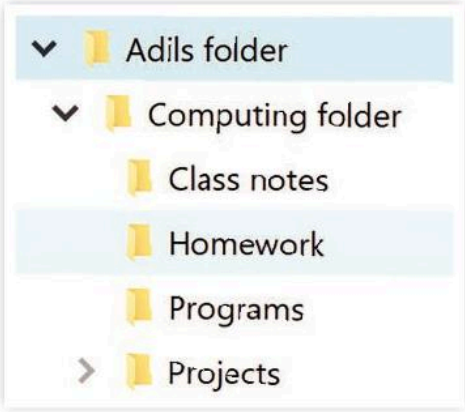
File Explorer gives information to help you find a file. The image shows a list of files and information that can help you identify them.



Using folders

Folders can help you to organise your work. You can organise your work by subject or type. For example, you can set up a folder for your computing work.

Inside the computing folder, you can set up folders to contain different types of work. For example, you could set up folders for homework, class notes, projects and programs.



How to create a folder

In File Explorer, create a new folder by following these steps.

1. Go to the area where you want to add a new folder.
2. Right click using your mouse.
3. Select New from the menu that appears, then click Folder.
4. Enter a name for the new folder.



Activity

Look at the list of files at the bottom of page 14. Sunny is searching for a file she knows she created using Word on 5 January 2019. Which files in the list can she dismiss? For each file, say why.



Extra challenge

Open your work area in File Explorer, either on a school computer or at home. Create a folder called 'Back-ups'. Copy a file to your 'Back-ups' folder.

Think again

How do you organise your data files at home and at school? What can you do to improve the way you use folders and file names?

1.6 Tips for saving files

In this lesson

You will learn:

- why it is important to save your work correctly
- some rules to help you remember to save your work correctly.

Why saving your work is important

Saving work correctly is important. Failing to save work correctly could mean:

- you will lose work and must start again
- you may not be able to find work when you need it again.

Rules to help you when you are saving files

Rule 1: Use a helpful file name

When you save a file, you must give it a name. Choose a name that describes the information in your file.

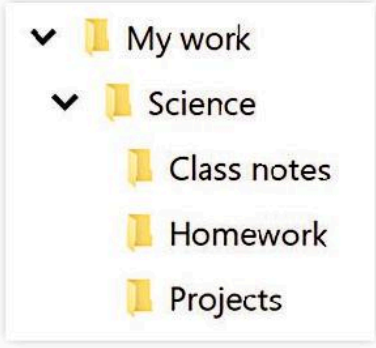
Example: You have just completed a science homework assignment on big cats. The assignment is to be handed in on 24 April 2020.
A good file name is:

- Science_homework_big_cats_24_April

Rule 2: Use folders to organise your work

Using folders to organise your files can help you to find your work. Folders also allow you to use shorter file names. You can create a folder to hold all your science work. Now your big cats file name can be shortened to:

- Homework_big_cats_24_April





Activity

There is a folder called 'Homework' shown in the directory structure. How can the file name 'Homework_big_cats_24_April' be shortened if the file is stored in that folder?

Rule 3: Save your work often

Get used to saving your work regularly and often. Don't wait until you have finished a piece of work before saving it. There will be times in between typing that you pause to think about what you will write next. That is a good time to save your work.

Rule 4: Be careful not to overwrite your files

Sometimes you will see a warning on screen like the one opposite. The warning tells you that you are about to replace a file you have previously saved. You might be about to lose important work.

If you are not sure what to do, cancel the Save and start again. If the problem continues, ask your teacher for help.



Rule 5: Make copies of important files

Sometimes you have a file that is so important that losing it would be a disaster. For example, it might be an important project that has taken you a long time to complete. You can reduce the risk of losing important work by making a copy of the file. The copy is called a back-up file.



Activity

Make an information sheet to promote good practice about saving files. Use the best practice rules in this section to help you.



Extra challenge

Work with a partner. How well does your partner use file names and folders? Suggest how your partner can make improvements to their file storage.



Explore more

Spend time at home with a family member who has work stored on a computer or tablet. How do they organise their data files? Can you suggest ways to improve how they organise their files? Can you learn anything from the way they work?